University of BRISTOL

The Effect of Forest Age on Machine Learning Evapotranspiration Generalisation

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Abstract (200 words)
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Key words: Evapotranspiration, Forest Age, Machine Learning, FLUXNET.
Word count: 4000 (Literature Review: 3000; Project Plan: 1000)

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science by advanced study in Bioinformatics in the Faculty of Health and Life Sciences.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
May 11, 2026

Contents

Literature Review

Introduction

Evapotranspiration

Terrestrial evapotranspiration (ET) is a key component of the global hydrological cycle describing the transition of water from liquid to gas at Earth's surface and further transfer to the atmosphere, and it is a crucial component of the energy and water exchange between terrestrial ecosystems and the climate system. The term emphasises the combination of water vapourisation flux from (1) abiotic evaporation from soil and plant canopy surfaces, and (2) biotic transpiration through stomata to the atmosphere (Katul et al. 2012). Approximately 60% of precipitation on the land surface is returned to the atmosphere through ET (Oki and Kanae 2006), and the associated latent heat flux, λET (where λ is the latent heat of vaporisation), accounts for 50% of the solar radiation absorbed by the land surface (Trenberth et al. 2009). ET can be generally described through:

$$ET = T + E_{Soil} + E_{Canopy} \tag{1}$$

where T refers to transpiration through plant stomata, E_{soil} to evaporation from the soil surface, and E_{Canopy} to evaporation from plant canopies (via rainfall interception) (Lawrence et al. 2007). The contribution of these variables to overall ET is not equal, with Lian et al. (2018) estimating a global T/ET ratio of 0.62±0.06, indicating plant transpiration as the leading component of ET worldwide. The partitioning of ET into E ($E_{soil} + E_{Canopy}$) and T within ecosystems is associated with water use efficiency (WUE) and has use in irrigation management, where T is linked with plant productivity, while E does not directly contribute to production (Kool et al. 2014). Accurate partitioning estimates are also important in aiding carbon cycle projections given the coupled effect of transpiration and land-atmosphere carbon dioxide exchanges through stomata.

Forest Complexity

The magnitude of ET is governed by many environmental factors such as solar radiation, leaf area index (LAI), air temperature, vapour pressure deficit (VPD), soil moisture, and wind speed (Yang et al. 2023). Of these, Lian et al. (2018) identified LAI as the primary driver of T/ET spatial patterns, with Kool et al. (2014) suggesting that in systems with full canopy cover ET is often assumed to be "similar enough to T to permit correlation of ET with biomass". However, this assumption may break down in densely vegetated systems with multi-layer canopies such as forests, despite high LAI. In these systems, saturation issues with the remote measurement of LAI lead to issues in capturing the vertical canopy complexity in full, leading to a loss of information which drives rainfall interception, E_{Canopy}, and subsequent ET dynamics (Q. Wang et al. 2022).

Hardiman et al. (2013) demonstrate that while LAI may be stable in stands older than 50 years, canopy complexity continues to increase with age through >160 years of stand development and had a greater impact on annual net primary productivity (NPP) and overall resource use efficiency than LAI across all stands, suggesting vegetation indices may lack the ability to fully explain ET partitioning in structurally complex systems. Oishi et al. (2020), in a chronosequence study on broadleaf forests in the US, report that interception of incoming rainfall strongly increases with age from ~2% (12 year old stand) to 20% (200 year old stand) despite only small differences in LAI, an effect in part due to branch architecture which varies with age. This study also reported that annual canopy transpiration (T) was over twice as high in the 35 year old stand compared to the 12 and 200 year old stands, with increasing canopy interception in the oldest stand largely offsetting the reduction in T such that total ET was similar between the two older stands, but remained lower in the youngest stand (Figure 1). Furthermore, Murakami et al. (2000) formed a model based on the Penman-Monteith (PM) equation, and this predicted an ET-stand age relationship in a mixed forest which peaks at 20 years. Murakami et al. (2000) go on to claim that this reflects an LAI-stand age relationship, however LAI has been shown to saturate over time and before canopy complexity peaks in forests (Hardiman et al. 2013), suggesting that this relationship may not hold true across different forest types. While LAI remains an important structural parameter for informing ET predictions and quantifying biological processes relating to vegetation dynamics, these findings together suggest that age-driven changes in canopy complexity and architecture introduce ET dynamics that may persist beyond LAI saturation.

Alongside canopy architecture, stomatal conductance (g_s) declines as trees grow taller due to gravitational reductions in leaf water potential and decreasing hydraulic conductance with increasing path length from soil to leaf (Ryan and Yoder 1997). For example, Schäfer et al. (2000) demonstrated that the mean stomatal conductance of tree crowns decreased by approximately 60% over a 30m increase in tree height, and Hubbard et al. (1999) established a 44% drop in hydraulic conductance in old trees compared to young trees in a mixed age stand. As g_s directly influences T (Monteith 1965), these age related changes introduce further complexity into ET dynamics and suggest that age could be a key source of variability in ET in forest ecosystems.

Forest Age

Forest age distributions are increasingly being affected by climate change-driven disturbances - such as fire, drought, insect attack, and pathogens - which are expected to continue in the coming decades (Solomon et al. 2009; Seidl et al. 2017). Some of these disturbances are able to completely change forest demography by causing widespread mortality (Dale et al. 2001), effectively resetting the age of the forest landscape. Land use changes also directly affect forest age distributions, for example the recovery of deforested areas in southern China following the implementation of forestation policies has resulted in a complex pattern of forest age structure in recent years, and an increase in average tree cover from 21% in 1987 to 38% in 2016, leading to large areas now being dominated by young forests (Tong et al. 2020; Leng et al. 2024), with similar patterns of young forests due to recent disturbance found worldwide (Besnard et al. 2021).

As forests are exhibiting greater age heterogeneity, and with climate change increasing the severity and frequency of disturbances worldwide, forest age may represent an increasingly important variable for ET estimation, capturing age-specific effects such as canopy complexity and hydraulic conductance limitations that persist beyond LAI saturation and may not be resolved by vegetation indices used by ET estimation models.

Evapotranspiration Prediction

Importance of ET Prediction

Estimating ET accurately is important in understanding hydrological processes, management of irrigation systems for effective water use in agriculture, prediction and mitigation of droughts, determining the adaptive mechanisms of vegetation, and in assessing the overarching patterns of global climatic shifts (Amani and Shafizadeh-Moghadam 2023; Y. Xue et al. 2025). These assessments can, in turn, steer regulations and policies in climate science and water resource management, representing a critical link between the quantification of terrestrial water fluxes and land systems governance (Haiyang Shi et al. 2024).

The direct measurement of ET is the most accurate and uses lysimeters, eddy covariance (EC), or sap flow, but these are costly to set up and only measure at the plot scale, resulting in spatially constrained measurements which makes them unsuitable for describing ET at larger ecosystem scales (Amani and Shafizadeh-Moghadam 2023; Liyew et al. 2025). The uneven distribution and low sampling rate of EC data in ecoregions of Africa, Oceania (excluding Australia), and South America may be attributed to the costs associated with these methods, despite efforts to mitigate this (Hill et al. 2017; Markwitz and Siebicke 2019). As a result of these issues, various methods to estimate ET indirectly in areas without the capacity for direct measurement have been derived, including: (1) traditional process-based models which rely on pre-defined equations such as the PM and Hargreaves-Samani (HS) equations (Monteith 1965; Hargreaves and Samani 1985); (2) remote sensing (RS) - based models, for example the SEBS model (Z. Su 2002); and more recently, (3) machine learning (ML) methods such as Artificial Neural Networks (ANN) and Random Forest (RF) models trained on historical data (T. Xu et al. 2018). The development of these methods has permitted global ET estimates to be produced using more easily acquired variables such as air temperature and relative humidity (S. Pan et al. 2020), yet significant uncertainties remain, particularly in underrepresented regions and ecosystems with sparse EC coverage, as well as in complex or heterogeneous landscapes (T. Xu et al. 2018).

ET Estimation Methods

There are many different methods for the estimation of ET as no single model performs reliably across all conditions and regions (Derardja et al. 2024), thus the data requirement can be significantly varied across models (Zhao et al. 2013). Process-based models define the relationship between input variables and ET explicitly for estimations which are developed through hydrological theory, while ML approaches do not explicitly define relationships, instead relying on the underlying architecture of models to optimise against the provided training data to derive the relationship between inputs and ET. RS data cannot provide direct estimations of ET, but instead estimates retrievable surface parameters associated with ET such as vegetation indices like LAI, surface temperatures, and soil moisture (Amani and Shafizadeh-Moghadam 2023).

Process-based methods such as the PM equation are able to achieve high accuracy across climatic zones, yet they suffer from a requirement of comprehensive input meteorological data and their rigid structure limits adaptability, for example their calibration is typically time and space dependent which restricts generalisation (Amani and Shafizadeh-Moghadam 2023). The continuous and broad spatial monitoring of variables affecting ET has been facilitated through advances in RS technology, yet RS methods are limited by low per pixel resolution (e.g $30m^2$ for Landsat and $500m^2$ for MODIS) which affects fine scale spatial prediction accuracy, alongside issues with temporal coverage from cloud interference and long revisitation times (16 days for Landsat) (J. Xue and B. Su 2017).

ML methods have been used extensively for ET estimation in recent years and have demonstrated high spatiotemporal prediction accuracy and a capability to outperform process-based and RS-based models (S. Pan et al. 2020; Yan et al. 2023). Their increased precision of ET estimations in data rich regions may be due to their ability to leverage both linear and non-linear relationships for mapping between input data and ET estimates (Y. Xue et al. 2025). In data limited contexts, coupled ML models which use process-based model outputs as engineered input features have shown improved performance compared to standalone methods. For example, Yan et al. (2023) deployed a HS-ML coupled model which incorporated the HS model output as an input feature for a long short-term memory neural network (LSTM) and demonstrated better performance compared to the standalone HS and ML methods in predicting daily potential ET (ET_0), while Y. Liu et al. (2021), in a reverse approach, used ML to predict g_s before applying this value to the PM equation and demonstrated greater performance over the PM equation alone. Traditional (non deep learning) ML approaches, such as RF and Support Vector Machine (SVM), are commonly used and also show good ET predictive performance, with T. Xu et al. (2018) comparing the capabilities of five ML models against EC data as ground truth, finding the RF model to slightly outperform the others in uncertainty, with all models predicting ET to high accuracy (R^2 0.87 – 0.89). LTSMs are a specialised form ANN which are highly capable in extracting long term dependencies within time series data and handling noisy datasets, indicating strong potential for broad application and precise estimations in ET using long-term EC datasets for training (Yan et al. 2023).

ML methods, while promising in reducing uncertainties and improving ET estimation accuracy, can struggle with extrapolation beyond training data and lack the interpretability of process-based models, often being termed 'black boxes' due to their inherent complexity and lack of decision making explainability (Hassija et al. 2024). The quality of ML ET estimations is highly dependent on the provided training data, model hyperparameter tuning, and choice of predictors (feature selection) - all of which relies on user knowledge and pre-existing data for the specific ecosystem in question (Amani and Shafizadeh-Moghadam 2023). Therefore careful consideration must be taken during ML training and evaluation to form a relevant and accurate ET estimation model for the specific use case.

Feature Selection and FLUXNET

The selection of features in ML ET estimation models can vary from a minimal data input (Ahmadi et al. 2024) to a broad choice of variables from multiple sources (Y. Liu et al. 2021). Table 2 presents examples on the range of input features included in ML models used for ET predictions. The range of selected input features is often constrained by the availability of meteorological data in the target region, with T. Xu et al. (2018) demonstrating that successively adding more input variables increased ET prediction accuracy and reduced uncertainty in five different ML models. However, the number of input variables used may also vary during attempts to form models which can be applied to data limited regions for greater generalisation power (Ahmadi et al. 2024). In either case, access to standardised, high quality data is key in forming ML models for ET.

An important and commonly used resource to form ML ET models is the FLUXNET (Baldocchi et al. 2001; Pastorello et al. 2020), which is formed of data collected at flux towers worldwide and comprises gas and energy flux measurements - including latent heat (λET) from which ET can be directly calculated, alongside meteorological and biological measurements at these sites. The EC technique uses these data to estimate land-atmosphere net exchange of flux variables at the ecosystem scale within a variable footprint around the tower. The standardisation and open access of FLUXNET data across sites enables confident comparisons and EC data access across multiple ecosystems (Pastorello et al. 2020), and yet despite the abundance of variables contained within the FLUXNET, forest specific features such as stand age are rarely included at sites. Given the potential impact of forest age on ET described above and the widespread use of FLUXNET data in training ML ET models (Jung et al. 2010; Simon Besnard et al. 2018; R. Wang et al. 2022), this omission represents a gap in the feature space available to these models which potentially limits their ability to generalise across forests of varying successional stages and structural complexities.

Machine Learning and Integrating Forest Age

Machine Learning for Age-ET Relationships

ML represents an avenue for integrating forest age as a variable for ET estimation models. ML methods have an ability to capture both linear and nonlinear relationships between predictor and target variables due to their inherent lack of prior assumptions between the two (R. Wang et al. 2022; Simon Besnard et al. 2018). Critically, this flexibility of ML models has previously been used to include forest age, with Simon Besnard et al. (2018) using a RF model with and without forest age as a predictive variable for net ecosystem productivity (NEP) estimations, finding strong support for a nonlinear relationship between NEP and forest age which demonstrates both the strength of this ML method in detecting these relationships and a capacity for including forest age as an input feature. This adaptability of ML methods to detect underlying patterns within complex datasets provides an avenue for the age-driven effects of forest age on ET estimations using FLUXNET data to be further explored.

Incorporating Forest Age

Vegetation age is not a standardised variable within the FLUXNET2015 dataset, with site level metadata varying considerably in the availability and precision of age-related information (Pastorello et al. 2020). This presents a difficult issue for investigating age-driven effects, particularly within ML models which require consistent and quantified inputs to derive relationships.

Simon Besnard et al. (2018) considered four different scenarios in databases and publications to define forest age across the FLUXNET: (1) Complete stand replacement events affecting the whole area under the flux tower footprint area, where this information is used to set the age; (2) Minor stand disturbance events, such as insect outbreaks or wind throws covering minor parts of the footprint, where age was not adjusted; (3) marginal logging events, where limited reporting of logging intensity or age of logged trees resulted in no changes to stand age; and (4) partial substantial disturbance where a disturbance event impacts the majority of trees, and here the age is considered the difference between the flux observation year and the year of disturbance. These four scenarios led to age being quantified at 126 FLUXNET sites, but the uncertainties involved with these ages are large due to the qualitative and non-standardised reporting of disturbances, with particularly large uncertainties at sites where ages were calculated from disturbances occurring before flux observations began. Despite these uncertainties, the resulting list of flux sites and quantified ages provides a suitable training dataset for ML models to be used in future studies involving FLUXNET and forest age.

The methods used by Simon Besnard et al. (2018) to determine site ages reflects the current challenges and uncertainties in investigating forest age with FLUXNET sites. Field-based surveys provide a route to expand existing FLUXNET site data by accurately characterising tree demography but are labour-intensive and therefore spatially restricted (Battison et al. 2024). Satellite-based RS approaches overcome this but lack sufficient resolution which limits their applicability with FLUXNET data due to significant differences in measurement scale.

The Global Canopy Atlas is a novel data source which utilises airborne laser scanning (ALS) acquisitions to create an analysis-ready map of canopy height and elevation covering 56,554km^2 at 1m^2 resolution across all major biomes, forming an accurate 3-dimensional map of canopy structures (Fischer et al. 2025). To showcase this atlas, Fischer et al. (2025) developed a framework to standardise forest turnover quantification, highlighting its potential application in automating the quantification of forest ages altered by disturbances from LiDAR imaging at a spatiotemporal resolution great enough to permit use with FLUXNET data.

The potential of standardised, high resolution, and large scale forest age surveys on the horizon opens up the possibility for ML ET models to include forest age as a spatially continuous input feature, which may enable models trained on FLUXNET data to generalise more accurately across forest age gradients at broad scales.

Model Generalisation

ET models can be calibrated or trained with local or regional data. However, local models may demonstrate strong predictive performance locally but weak performance at other locations, potentially rendering application outside of the training area unusable. Therefore, for ML models trained on specific flux tower data, it is important to assess the generalisation power against towers excluded from the training data. ML models trained with data from multiple towers or across a regional scale can be useful for predictions at data-limited sites within a particular region or ecosystem, and have demonstrated more stable performance across multiple sites, albeit with slightly lower accuracy in ecosystems well represented by locally-trained models (Ferreira et al. 2021).

The limited and uneven spatial distribution of flux tower sites in the FLUXNET highlights the necessity for robust models which are able to extend ET estimations across a diverse range of landscapes and climatic conditions. Models with better generalisation capabilities should improve estimations for ET in regions with lower sampling rates of EC, in turn improving assessments of local climatic shifts and guiding efficient water use practices for agriculture in areas of lower income (Haiyang Shi et al. 2024). Given the potential impact of these ET estimations in steering regulations and policies in climate science and water resource management, understanding how the inclusion of new variables such as forest age affects model generalisation is crucial before these models can be confidently applied at broader scales.

Summary

These findings suggest that the age of a forest represents an underused but potentially important variable in ET estimation models, particularly in dense and closed-canopy systems. Age-driven processes such as canopy architecture, rainfall interception, and changes to hydraulic and stomatal conductance introduce complexity to ET dynamics which may not be fully captured by indices used in contemporary models. As forest age distributions become increasingly heterogeneous, the omission of forest age as an input variable for ET models may be limiting the capacity for these models to generalise across successional forest stages.

This project will attempt to quantify the effects of including forest age as a predictor on the generalisation of ET models trained on FLUXNET forest sites, using ML models for their ability to capture both linear and nonlinear effects across large datasets. Crucially, improved predictions are only meaningful when grounded in the biophysical processes underpinning ET and used to improve our understanding of how these may shift under increasing disturbances such as climate change. Without this context, model prediction accuracy is constrained to a technical metric with limited ecological relevance.

Project Aims and Objectives (500 words)

Aims

- Overview of how this project will answer the question/area proposed at end of introduction.

Project Plan (500 words)

Data Acquisition

Site Selection

Analysis

Feature Engineering

ML Model Setup and Training

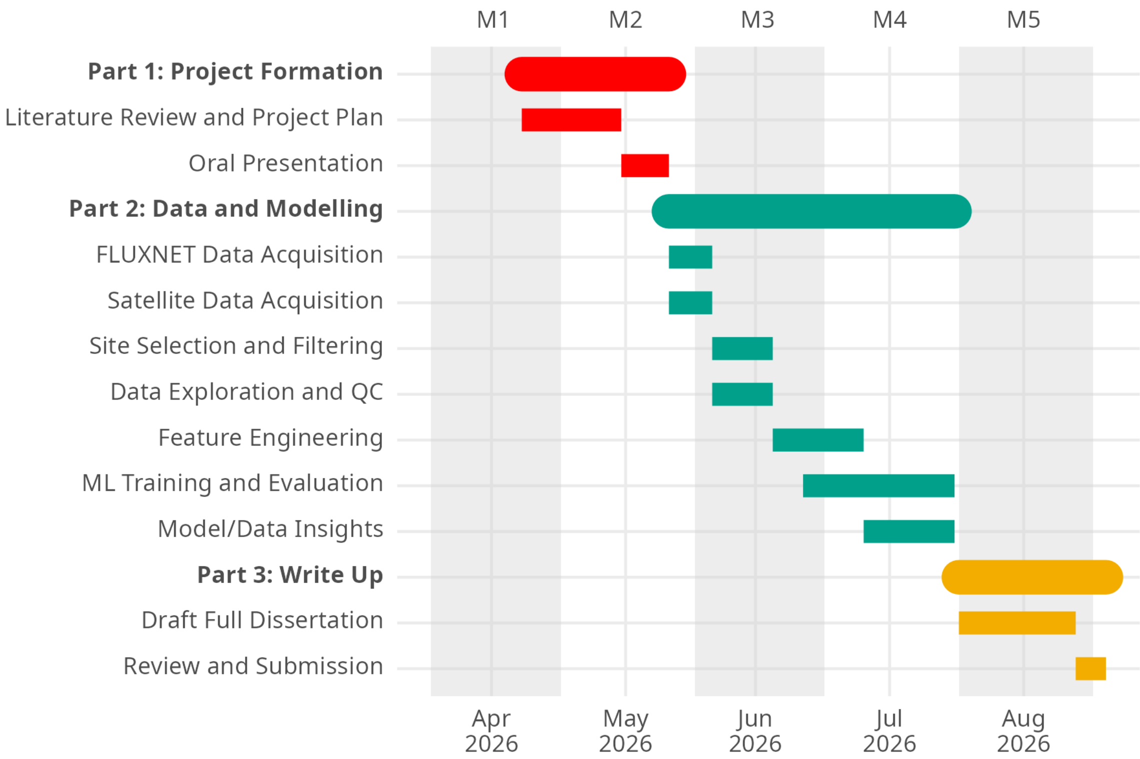
ML Model Evaluation

ML Model Predictions

Tools

Interpretation

Gantt Chart



M1
M2
M3
M4
M5
Part 1: Project Formation
Literature Review and Project Plan
Oral Presentation
Part 2: Data and Modelling
FLUXNET Data Acquisition
Satellite Data Acquisition
Site Selection and Filtering
Data Exploration and QC
Feature Engineering
ML Training and Evaluation
Model/Data Insights
Part 3: Write Up
Draft Full Dissertation
Review and Submission
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026

Risk Register (~ 250 words)

Table 1: Risk register

Nature of the risk	Likelihood and potential impact on the project	Mitigation strategy

References

Ahmadi, Arman et al. (2024). "SolarET: A generalizable machine learning approach to estimate reference evapotranspiration from solar radiation". In: *Agricultural Water Management* 295, p. 108779.

Amani, Shima and Hossein Shafizadeh-Moghadam (2023). "A review of machine learning models and influential factors for estimating evapotranspiration using remote sensing and ground-based data". In: *Agricultural water management* 284, p. 108324.

Baldocchi, Dennis et al. (2001). "FLUXNET: A new tool to study the temporal and spatial variability of ecosystem-scale carbon dioxide, water vapor, and energy flux densities". In: *Bulletin of the American Meteorological society* 82.11, pp. 2415–2434.

Battison, Robin et al. (2024). "Tracking tree demography and forest dynamics at scale using remote sensing". In: *New Phytologist* 244.6, pp. 2251–2266.

Besnard, S et al. (2021). *Mapping global forest age from forest inventories, biomass and climate data, Earth Syst. Sci. Data, 13, 4881–4896.*

Besnard, Simon et al. (2018). "Quantifying the effect of forest age in annual net forest carbon balance". In: *Environmental Research Letters* 13.12, p. 124018.

Dale, Virginia H et al. (2001). "Climate change and forest disturbances: climate change can affect forests by altering the frequency, intensity, duration, and timing of fire, drought, introduced species, insect and pathogen outbreaks, hurricanes, windstorms, ice storms, or landslides". In: *BioScience* 51.9, pp. 723–734.

Derardja, Bilal et al. (2024). "Advancements in remote sensing for evapotranspiration estimation: a comprehensive review of temperature-based models". In: *Remote Sensing* 16.11, p. 1927.

Ferreira, Lucas B et al. (2021). "Generalizability of machine learning models and empirical equations for the estimation of reference evapotranspiration from temperature in a semiarid region". In: *Anais da Academia Brasileira de Ciências* 93, e20200304.

Fischer, Fabian Jörg et al. (2025). "The Global Canopy Atlas: analysis-ready maps of 3D structure for the world's woody ecosystems". In: *bioRxiv*, pp. 2025–08.

Hardiman, Brady S et al. (2013). "Maintaining high rates of carbon storage in old forests: A mechanism linking canopy structure to forest function". In: *Forest Ecology and Management* 298, pp. 111–119.

Hargreaves, George H and Zohrab A Samani (1985). "Reference crop evapotranspiration from temperature". In: *Applied engineering in agriculture* 1.2, pp. 96–99.

Hassija, Vikas et al. (2024). "Interpreting black-box models: a review on explainable artificial intelligence". In: *Cognitive Computation* 16.1, pp. 45–74.

Hill, Timothy, Melanie Chocholek, and Robert Clement (2017). "The case for increasing the statistical power of eddy covariance ecosystem studies: why, where and how?" In: *Global Change Biology* 23.6, pp. 2154–2165.

Hubbard, Robert M, Barbara J Bond, and Michael G Ryan (1999). "Evidence that hydraulic conductance limits photosynthesis in old Pinus ponderosa trees". In: *Tree physiology* 19.3, pp. 165–172.

Jung, Martin et al. (2010). "Recent decline in the global land evapotranspiration trend due to limited moisture supply". In: *Nature* 467.7318, pp. 951–954.

Katul, Gabriel G et al. (2012). "Evapotranspiration: A process driving mass transport and energy exchange in the soil-plant-atmosphere-climate system". In: *Reviews of Geophysics* 50.3.

Kool, Dilia et al. (2014). "A review of approaches for evapotranspiration partitioning". In: *Agricultural and forest meteorology* 184, pp. 56–70.

Lawrence, David M et al. (2007). "The partitioning of evapotranspiration into transpiration, soil evaporation, and canopy evaporation in a GCM: Impacts on land–atmosphere interaction". In: *Journal of Hydrometeorology* 8.4, pp. 862–880.

Leng, Yi et al. (2024). "Forest aging limits future carbon sink in China". In: *One Earth* 7.5, pp. 822–834.

Lian, Xu et al. (2018). "Partitioning global land evapotranspiration using CMIP5 models constrained by observations". In: *Nature Climate Change* 8.7, pp. 640–646.

Liu, Yan et al. (2021). "Assessment and comparison of six machine learning models in estimating evapotranspiration over croplands using remote sensing and meteorological factors". In: *Remote Sensing* 13.19, p. 3838.

Liyew, Chalachew Muluken et al. (2025). "A review of feature selection methods for actual evapotranspiration prediction". In: *Artificial Intelligence Review* 58.10, p. 292.

Markwitz, Christian and Lukas Siebicke (2019). "Low-cost eddy covariance: a case study of evapotranspiration over agroforestry in Germany". In: *Atmospheric measurement techniques* 12.9, pp. 4677–4696.

Monteith, John Lennox (1965). "Evaporation and environment". In.

Murakami, S et al. (2000). "Variation of evapotranspiration with stand age and climate in a small Japanese forested catchment". In: *Journal of Hydrology* 227.1-4, pp. 114–127.

Oishi, AC et al. (2020). “Quantifying the effects of stand age on components of forest evapotranspiration”. In: *Acta Horticulturae* 1300, pp. 89–96.

Oki, Taikan and Shinjiro Kanae (2006). “Global hydrological cycles and world water resources”. In: *science* 313.5790, pp. 1068–1072.

Pan, Shufen et al. (2020). “Evaluation of global terrestrial evapotranspiration using state-of-the-art approaches in remote sensing, machine learning and land surface modeling”. In: *Hydrology and earth system sciences* 24.3, pp. 1485–1509.

Pastorello, Gilberto et al. (2020). “The FLUXNET2015 dataset and the ONEFlux processing pipeline for eddy covariance data”. In: *Scientific data* 7.1, p. 225.

Ryan, Michael G and Barbara J Yoder (1997). “Hydraulic limits to tree height and tree growth”. In: *Bioscience* 47.4, pp. 235–242.

Schäfer, KVR, Ram Oren, and JD Tenhunen (2000). “The effect of tree height on crown level stomatal conductance”. In: *Plant, Cell & Environment* 23.4, pp. 365–375.

Seidl, Rupert et al. (2017). “Forest disturbances under climate change”. In: *Nature climate change* 7.6, pp. 395–402.

Shi, Haiyang et al. (2024). “Machine learning-based investigation of forest evapotranspiration, net ecosystem productivity, water use efficiency and their climate controls at meteorological station level”. In: *Journal of Hydrology* 641, p. 131811.

Solomon, Susan et al. (2009). “Irreversible climate change due to carbon dioxide emissions”. In: *Proceedings of the national academy of sciences* 106.6, pp. 1704–1709.

Su, Zhongbo (2002). “The Surface Energy Balance System (SEBS) for estimation of turbulent heat fluxes”. In: *Hydrology and earth system sciences* 6.1, pp. 85–100.

Tong, Xiaowei et al. (2020). “Forest management in southern China generates short term extensive carbon sequestration”. In: *Nature communications* 11.1, p. 129.

Trenberth, Kevin E, John T Fasullo, and Jeffrey Kiehl (2009). “Earth’s global energy budget”. In: *Bulletin of the american meteorological society* 90.3, pp. 311–324.

Wang, Quan et al. (2022). “Combining both spectral and textural indices for alleviating saturation problem in forest LAI estimation using Sentinel-2 data”. In: *Geocarto International* 37.25, pp. 10511–10531.

Wang, Ren et al. (2022). “Recent increase in the observation-derived land evapotranspiration due to global warming”. In: *Environmental Research Letters* 17.2, p. 024020.

Xu, Tongren et al. (2018). “Evaluating different machine learning methods for upscaling evapotranspiration from flux towers to the regional scale”. In: *Journal of Geophysical Research: Atmospheres* 123.16, pp. 8674–8690.

Xue, Jinru and Baofeng Su (2017). “Significant remote sensing vegetation indices: A review of developments and applications”. In: *Journal of sensors* 2017.1, p. 1353691.

Xue, Yayong et al. (2025). “A review of evapotranspiration estimation models: advances and future development”. In: *Water Resources Management* 39.8, pp. 3641–3657.

Yan, Xiaohui et al. (2023). “Deep learning for daily potential evapotranspiration using a HS-LSTM approach”. In: *Atmospheric Research* 292, p. 106856.

Yang, Yuting et al. (2023). “Evapotranspiration on a greening Earth”. In: *Nature Reviews Earth & Environment* 4.9, pp. 626–641.

Zhao, Lingling et al. (2013). “Evapotranspiration estimation methods in hydrological models”. In: *Journal of Geographical Sciences* 23.2, pp. 359–369.

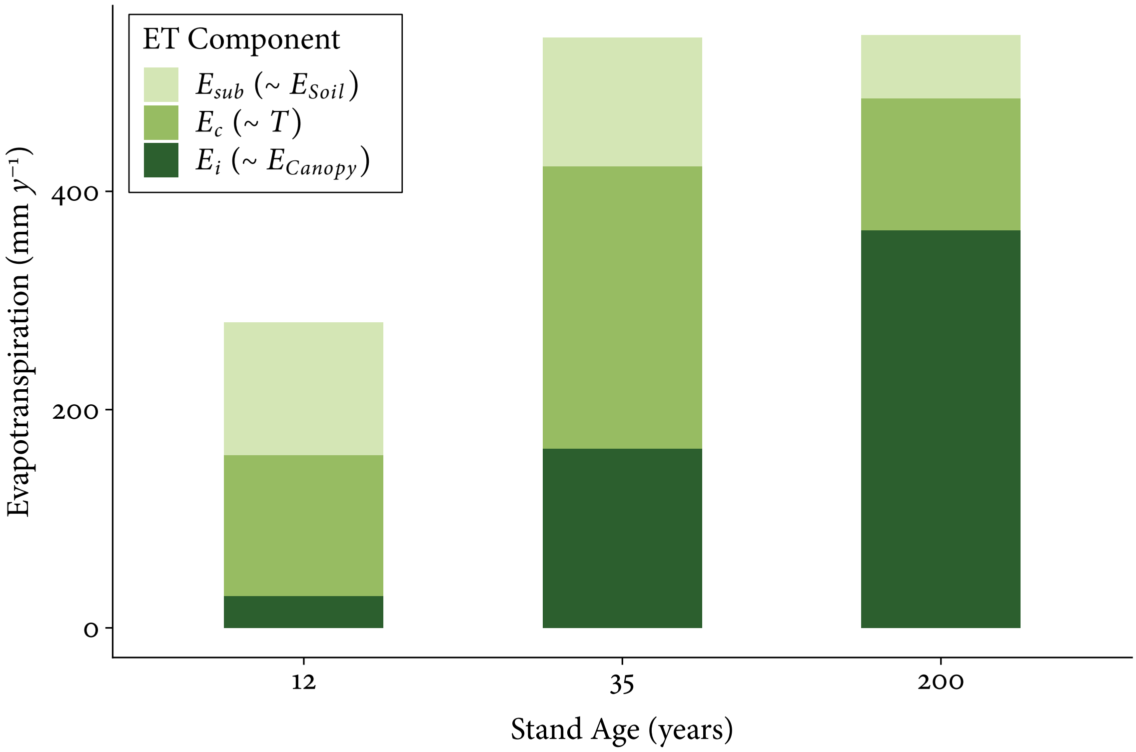


Figure 1: Components of annual evapotranspiration by stand age, redrawn from Oishi et al. (2020). Sub canopy evapotranspiration (E_{sub}), canopy transpiration (E_c), and canopy interception (E_i) correspond approximately to E_{Soil}, T, and E_{Canopy} in Equation 1, with E_{sub} additionally capturing under canopy transpiration.

Table 2: Feature selection of ML-based models in the literature

Primary Model(s)	**Training Target**	**Model Input Features**	**Ecosystem**
Random Forest, ANN, cubist, SVM, DBN (T. Xu et al. 2018).	Daily ET from 36 EC flux tower sites.	R_s, LAI, RH, T_a (daily averages). P (cumulative prior 30 days).	Mixed: grassland, cropland, ENF, BF, shrubland, wetland, barren land (Heihe River Basin, NW China).
HS-LSTM (Yan et al. 2023).	Daily ET_o calculated from FAO-56 PM as ground truth.	$T_{a,max}$ and $T_{a,min}$ (for ET_o from HS which is used as LTSM model input.	Not specified: Semi-humid zone with temperate continental climate (Songliao Basin, NE China).
XGboost, ANN (Y. Liu et al. 2021).	Daily ET from 17 flux tower sites.	T_a, P, C_a, SW, VPD, NDVI (daily averages). EVI, NIRv, SWIR (time between measurements not defined).	Cropland only.
CatBoost (Ahmadi et al. 2024)	Daily ET_o calculated from FAO-56 PM as ground truth.	R_s (hourly)	Mixed: 17 climatic zones including coastal, central valley, and desert (California, USA).

Where: R_s = solar radiation, LAI = leaf area index, P = precipitation, RH = relative humidity, T_a = air temperature, ENF = evergreen needle leaf forest, BF = broadleaf forest, $T_{a,max/a,min}$ = daily maximum/minimum air temperatures, C_a = atmospheric CO_2 concentration, SW = shortwave solar radiation, NDVI = normalised difference vegetation index, EVI = enhanced vegetation index, NIRv = near infrared reflectance of vegetation, SWIR = shortwave infrared band.